Akoizo Original Practice - Form 4 - Answer key

Division B | Elastic Launched Gliders | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A glider loops at launch. What is a first fix?

Unit: Trim: the transition from climb to glide.
Worked-case data: A glider climbs well but loops over at the top and then dives. Suggest a fix.
Separate investigation observations:
Launch at 0° bank: Glider loops over and dives; 8 s.
Launch at 30° bank: Climbs in a curve, rolls level; 22 s.
Launch at 70° bank: Climbs low and spirals; 12 s.

A. Pull farther

B. Move CG back

C. Launch with more bank

D. Add more decalage

Correct: C

2. (2 points) Two gliders have the same glide ratio. Which stays up longer from the same height?

A. They are equal

B. The heavier one always

C. The one flying more slowly

D. The faster one

Correct: C

3. (2 points) How many test flights must a complete log include?

Unit: The flight log and competition day.
Worked-case data: A team has five official flights in 6 minutes. After a trim flight, it plans official flights. Its first three are 21, 23, and 8 s (a beam hit). How should it use the remaining two?
Separate investigation observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

A. 10 or more

B. 3

C. 5

D. 20

Correct: A

4. (2 points) A glider loses a ballast piece during flight, dropping to 3.3 g. What is the issue?

A. Mass must stay within limits throughout the flight

B. No issue

C. Only check-in mass counts

D. The flight is doubled

Correct: A

5. (4 points) Why do very light gliders not always climb highest?

Unit: Launch energy: from stretched elastic to height.
Worked-case data: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.
Separate investigation observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

answer: They reach high speed but lose energy quickly to drag; heavier gliders carry momentum higher. (4 points).

6. (4 points) Why may teams bring their own ballast to check-in?

Unit: Rules, scoring, and planning five flights.
Worked-case data: A team with a complete log flies official flights of 22.4, 18.0, 25.1, 9.6, and 24.0 s. What is the final score and tiebreaker?
Separate investigation observations:
Glider A: Flights: 35, 12, 11, 10, 9 s.
Glider B: Flights: 24, 23, 22, 21, 20 s.
Both: Complete flight log.

answer: Scales differ, so ballast lets teams adjust mass to stay inside the legal window on the supervisor's scale. (4 points).

7. (4 points) Why rehearse a launch routine?

Unit: Launch technique and the ceiling.
Worked-case data: In practice, a 35 cm pull reaches 9 m. Competition ceiling is 7.5 m. Estimate a starting pull, assuming height scales with energy.
Separate investigation observations:
Practice room: 12 m ceiling; best pull 40 cm.
Competition room: 6 m ceiling with beams at 5.5 m.
First trim flight: 40 cm pull; glider hit a beam.

answer: A practiced sequence makes each launch the same, giving consistent flights under pressure. (4 points).

8. (4 points) What does a tightening spiral indicate?

Unit: Trim: the transition from climb to glide.
Worked-case data: A glider climbs well but loops over at the top and then dives. Suggest a fix.
Separate investigation observations:
Launch at 0° bank: Glider loops over and dives; 8 s.
Launch at 30° bank: Climbs in a curve, rolls level; 22 s.
Launch at 70° bank: Climbs low and spirals; 12 s.

answer: Too much turn or too little roll stability; reduce the turn or add dihedral. (4 points).

9. (4 points) How can a team increase glide ratio?

Unit: Glide physics: sink rate and glide ratio.
Worked-case data: A glider reaches 9.0 m and glides at 3.2 m/s with glide ratio 8. Estimate glide time.
Separate investigation observations:
Glider A: Peak 10 m; sink 0.55 m/s.
Glider B: Peak 8 m; sink 0.35 m/s.
Question: Which glides longer, ignoring climb time?

answer: Use long, thin wings, smooth surfaces, a clean fuselage, and good trim at the best angle of attack. (4 points).

10. (4 points) What should teams do with competition flight data?

Unit: The flight log and competition day.
Worked-case data: A team has five official flights in 6 minutes. After a trim flight, it plans official flights. Its first three are 21, 23, and 8 s (a beam hit). How should it use the remaining two?
Separate investigation observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

answer: Add it to the flight log to improve predictions for the next room and tournament. (4 points).

11. (6 points) Solve this case and explain your reasoning: A glider weighs 3.4 g and its CG is behind the target. What should the team do?

reasoning: It is below the 3.5 g minimum; add at least 0.2 g for margin.
Place that ballast in the nose to move the CG forward.
Recheck mass, balance, and hand glide. (3 points).

conclusion: Add about 0.2-0.3 g of nose ballast to fix both problems. (3 points).

12. (2 points) Which pull should the team use in this room?

Supplied synthetic observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

A. 45 cm

B. About 35 cm

C. 25 cm

Correct: B

13. (2 points) Define “Flight Period” in the context of Elastic Launched Gliders.

answer: 6 minutes beginning when the launch handle is returned. (2 points).

14. (2 points) Define “Launch bank” in the context of Elastic Launched Gliders.

answer: Wing tilt at release. (2 points).

15. (2 points) Define “Neutral point” in the context of Elastic Launched Gliders.

answer: CG position where pitch stability vanishes. (2 points).

16. (2 points) Define “Lift-to-drag ratio (L/D)” in the context of Elastic Launched Gliders.

answer: Measure of aerodynamic efficiency. (2 points).

17. (2 points) Define “Additional parameter” in the context of Elastic Launched Gliders.

answer: At least one extra, e.g., launch pitch or roll. (2 points).

18. (2 points) Define “Ballast” in the context of Elastic Launched Gliders.

answer: Added mass to set balance and mass. (2 points).